

Supplemental Material

Appendix-A

Statistical Description of Data

Wereldmuseum

Table 1: Wereldmuseum Knowledge Graph Description.

	count	graph len
objects	976,000	31,571,795
constituents [E21_Person]	39,619	769,315
constituents [E21_Person, E39_Actor, E74_Group]	51909	2307945
exhibitions	624	22,765
historical events	70	77,235
sites or themes	438	38,390

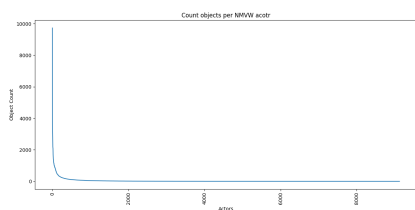


Figure 1: Objects per acotr distribution

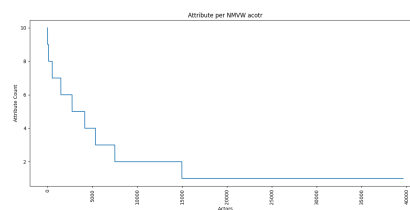


Figure 2: Available number of Attribute per actor distribution

Bronbeek Museum

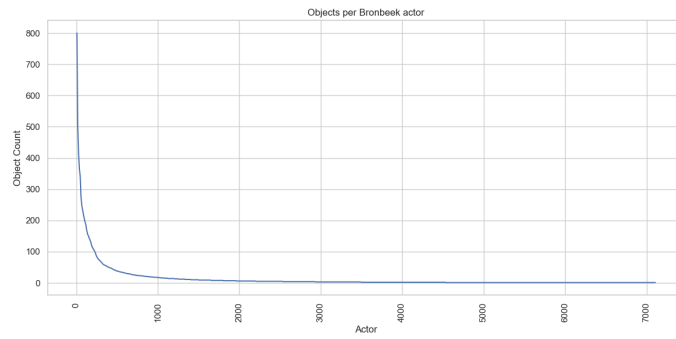


Figure 3: Object per actor distribution

Appendix-B

Defining Knowledge Discovery Task

The details steps of the conversion from the elicited Competency question to the question answered by the SPARQL Query.

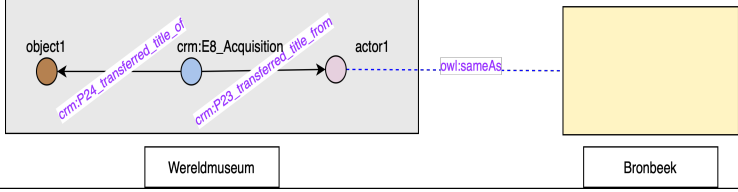
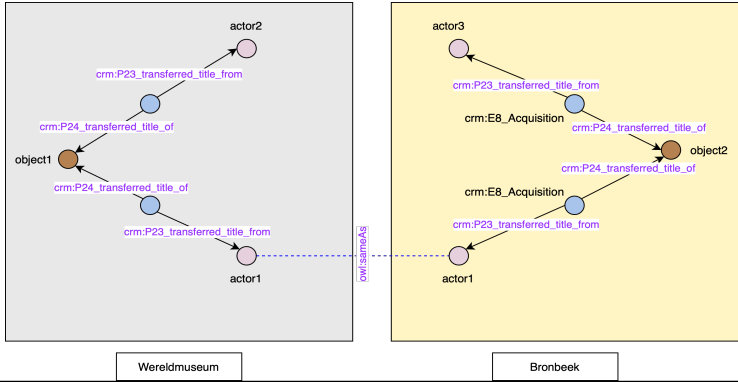
Table 2: Competency Questions and its interpretation for Query

	Original competency question(CQ)	Com-Ques-	Instance of CQ based on dataset	Information retrieval question to support CQ	Query used to report result (ΔH_{CQ})
CQ-1	Which persons were involved in the provenance of this object?		Which persons were involved in the provenance of these objects and have a military background?	How many actors do we have with one or more objects who also have a military background?	How many actors are involved in Wereldmuseum objects' acquisition or production events and also appear in the Bronbeek dataset?
CQ-1b			Is there a possible relation among objects as they are collected by the same actor?	How many objects have links with at least another object, based on the fact that they are collected by the same person?	How many objects are there from the Bronbeek dataset that are collected or produced by a Wereldmuseum actor, who has a connection to at least one other object?
CQ-2	Which objects are collected by person A?		Which objects are collected by person(s) with a military background?	How many objects are collected by person(s) with a military background?	How many actors are there whose acquisition or production actors also appear in the Bronbeek dataset?
CQ-3	Is there a relationship between person A and person B?		Is there a relationship between person A and person B through object collection events?	How many unique pairs of actors can be linked via the same objects' acquisition events?	How many new pairs of actors can be formed by connecting actors through object acquisition events?
CQ-4	Which objects were collected in this geographical location?		If the objects' production location is not known, for how many objects can we project a potential geographic location?	How many objects can we link to a location, based on their acquisition or production actor having at least one object whose production location is known?	How many Bronbeek objects are connected to Wereldmuseum actors, where at least one (Wereldmuseum) object per actor has a known production location?
CQ-5	Which objects were collected during this (historical) event?		Which objects may have a connection with a historical event, based on the fact that they are connected with some actors whose other object(s) have connections with a historical event?	How many objects have a connection with a historical event?	How many Bronbeek objects have connections with Wereldmuseum actors, where at least one (Wereldmuseum) object per actor has a connection with a historical event?
CQ-6a	Which objects were collected in this geographical location during this time period?		Which objects were collected in this geographical location during this time period from both datasets?	How many objects can be connected with other objects based on their being collected during a similar time by the same actor?	How many Bronbeek objects were collected by Wereldmuseum actors and have a similar acquisition time as some of their Wereldmuseum objects?
CQ-6b				How many objects can we possibly project the time of acquisition for based on their being collected by the same actor? (where we do not know the acquisition date, but we do know the acquisition date for some other object by the same actor)	How many Bronbeek objects have unknown acquisition dates and are connected to Wereldmuseum actors, where at least one (Wereldmuseum) object per actor has a known acquisition time?
CQ-6c				How many objects can be connected with other objects based on their being collected during a similar time and from a similar location by the same actor?	How many Bronbeek objects have connections with Wereldmuseum actors, where at least one (Wereldmuseum) object per actor has a connection with a historical event and the acquisition date is similar (+/- 20 years) to other objects from the same actor?

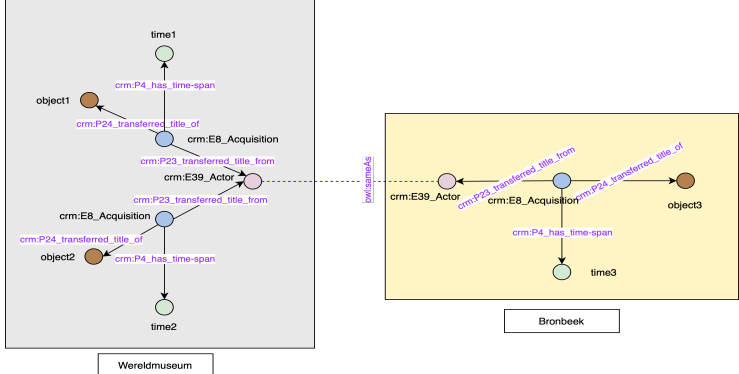
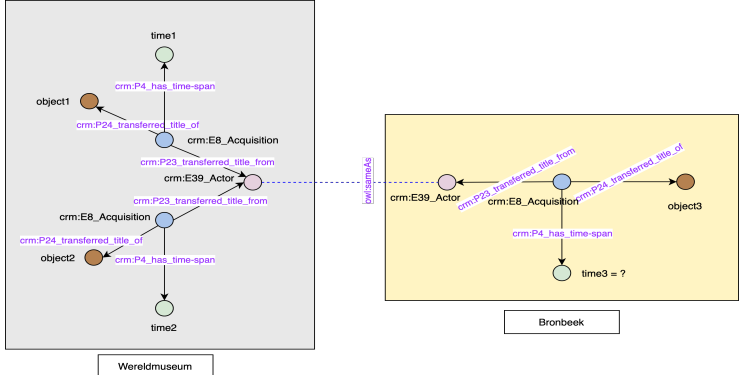
Appendix-C

SPARQL query Visual Illustration for each CQ

	Competency Question	Query in Natural Language
CQ-1a	Which persons were involved in the provenance of these objects and have a military background?	How many actors are involved in Wereldmuseum objects' acquisition or production events and also appear in the Bronbeek dataset?
	The diagram illustrates the provenance of object1. It shows a sequence of events: object1 is transferred to a person (blue circle) via 'crm:P24_transferred_title_of', who then acquires it ('crm:E8_Acquisition') and transfers it to actor1 ('crm:P23_transferred_title_from'). Actor1 is linked via a dashed line labeled 'owl:sameAs' to a yellow box representing the Bronbeek dataset. The Wereldmuseum dataset is shown as a grey box containing the initial part of the chain.	
	Answer: actor1	
CQ-1b	Is there a possible relation among objects as they are collected by the same actor?	How many objects are there who are collected or produced by an actor, who has connection to at least two objects and the actor is not “unknown”?
	The diagram shows two separate provenance chains. On the left (Wereldmuseum, grey box), actor1 (pink circle) is connected to object1 and object2 via 'crm:P23_transferred_title_from' and 'crm:P23_transferred_title_to' respectively. On the right (Bronbeek, yellow box), actor2 (pink circle) is connected to object3 and object4 via 'crm:P23_transferred_title_from' and 'crm:P23_transferred_title_to' respectively. A dashed line labeled 'owl:sameAs' connects actor1 and actor2, indicating they are the same person. Both actors are also connected to their respective objects via 'crm:E8_Acquisition' and 'crm:P24_transferred_title_of'.	
	Answer: Before Link $\rightarrow \{object1, Object2\}$	

	After Link $\rightarrow \{object1, Object3\}, \{object2, Object3\}, \{object2, Object4\}$	
CQ-2	Which objects are collected by person(s) with a military background?	How many objects are there whose acquisition or production actor(s) also appears in the Bronbeek dataset?
		
	Answer: After Link: Object1	
CQ-3	Is there a relationship between person A and person B?	How many (unique) pairs of actors are connected through objects' acquisition event(s)?
		
	Answer: Before Link $\rightarrow \{actor1, actor2\}$ After Link $\rightarrow \{actor1, actor3\}, \{actor2, actor3\}$	
CQ-4	If the objects' production location is not known, for how many objects can we project a potential geographic location?	How many objects that are connected to actors where at least one object per actor, we know their production location. Or, how many objects can we relate to a geographic location?

	Wereldmuseum	Bronbeek
	Before Link $Object1 \rightarrow relatesto \rightarrow Place1$ $Object2 \rightarrow relatesto \rightarrow Place1; Count : 2$ Answer: After Link $Object3 \rightarrow relatesto \rightarrow Place1; Count : 3$	
CQ-5	Which objects have connections with historical events?	How many object may have connection with historical event, based on the fact that, they are connected with some actors, whose other object(s) has connection with historical event?
	Wereldmuseum	Bronbeek
	Before Link $object1 \rightarrow relatedto \rightarrow historischevent1; Count1$ $Object2 \rightarrow relatedto \rightarrow Place1; Count : 2$ After Link $object2 \rightarrow relatedto \rightarrow historischevent1; Count2$	
CQ-6a	Which objects were collected in this geographical location during this time period from both dataset?	How many objects can be connected with one other object based on they are collected during similar time and by the same actor?

	 Wereldmuseum Bronbeek	
	Before Link: $\{object1, object2\}, if time1 = time2; Count : 1$ After Link $\{\{object1, object3\}, \{object2, object3\}\};$ if $time1 = time2 = time3; Count : 3$	
CQ-6b	Which objects were collected in this geographical location during this time period from both dataset?	How many objects can we possibly project the time of acquisition based on if they are collected by the same actor?
	 Wereldmuseum Bronbeek	
	Before Link: $time(Object3) = ?; Count : 0$ After Link $time(Object3) = time1 OR time2; Count : 1$	
CQ-6b	Which objects were collected in this geographical location during this time period from both dataset?	For How many objects can we suggest possible location(s) based on they are collected by the same actor and in a similar time-period?

